



W1 COS30045 Data Visualisation

Lecture

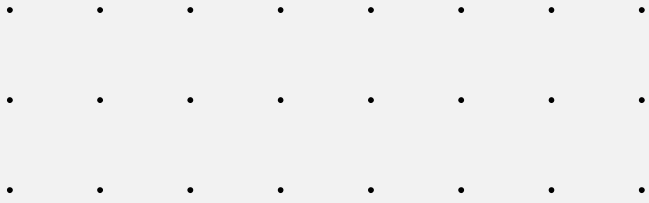
Class

Agenda

- 1.0 Introduction to unit
- 1.1 Intro to data visualisation
- 1.2 Data processing concepts
- 1.3 Data processing using KNIME

Agenda

- Finding and reviewing data
- Data cleaning
- Dealing with missing data
- Data transformation

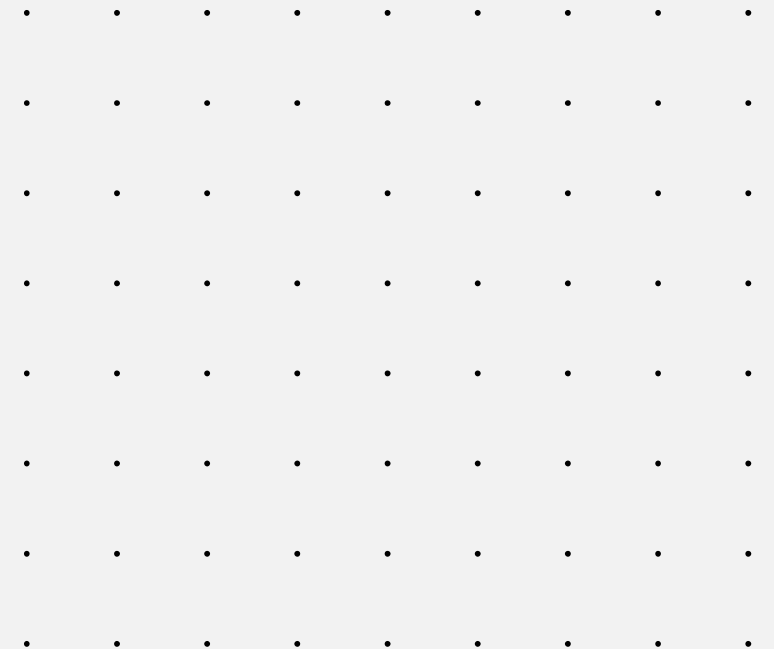


W1.0 COS30045

Lecture

Agenda

- Introduction to COS30045



Introduction to Unit: Staff Contacts

Convenor and Lecturer: Dr Karola von Baggo

See Canvas Home Page for:

- Contact details
- Book appointment



Tutors: Various...

See Canvas Home Page for:

- Contact details
- Book appointment



Canvas Learning Management System

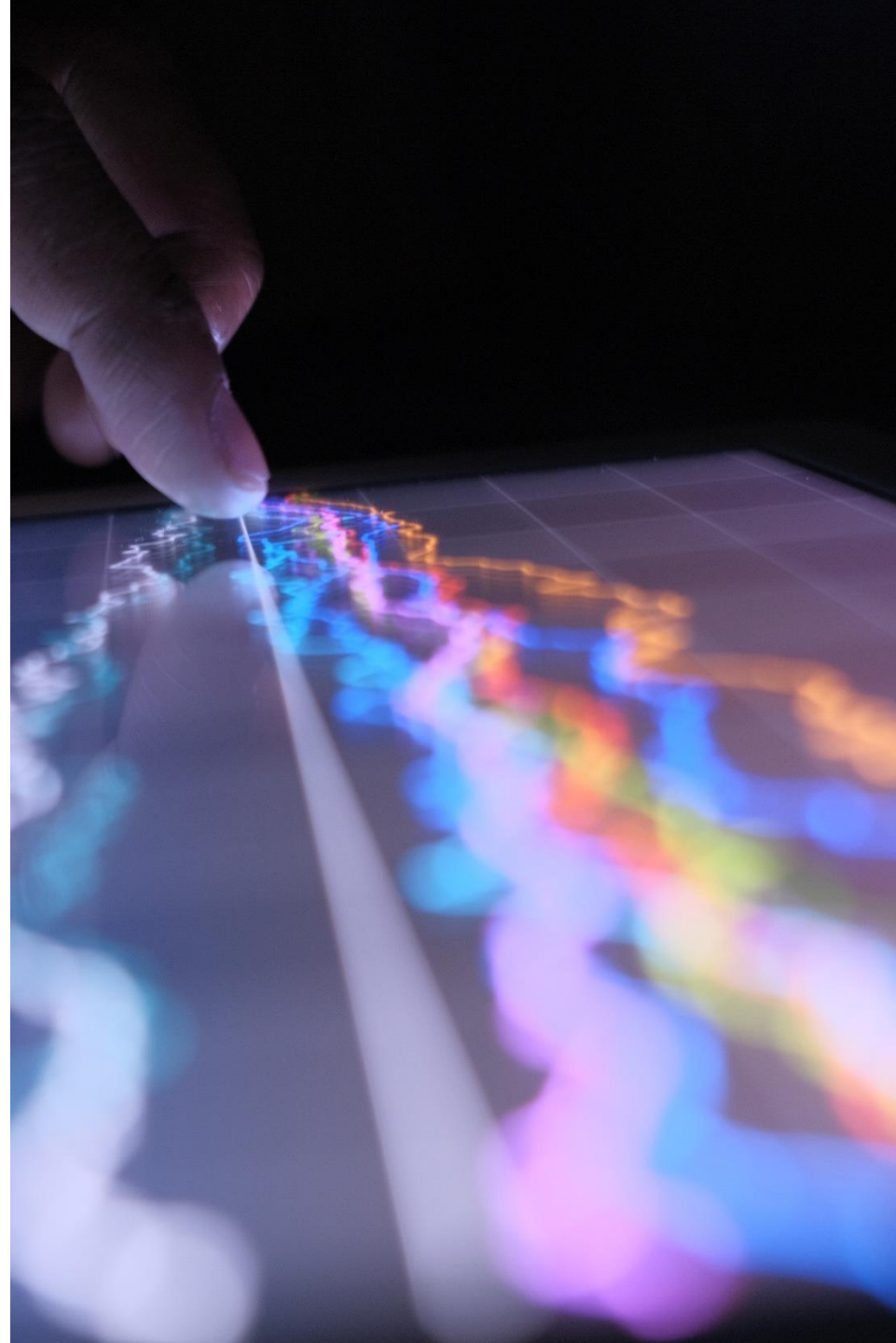
Things to look out for:

- Unit Outline (Syllabus)
- Learning Materials (Modules)
- Assessment Details and Submission points (Assignments)
- Discussion Board (Discussions)
- Contacts (Home)



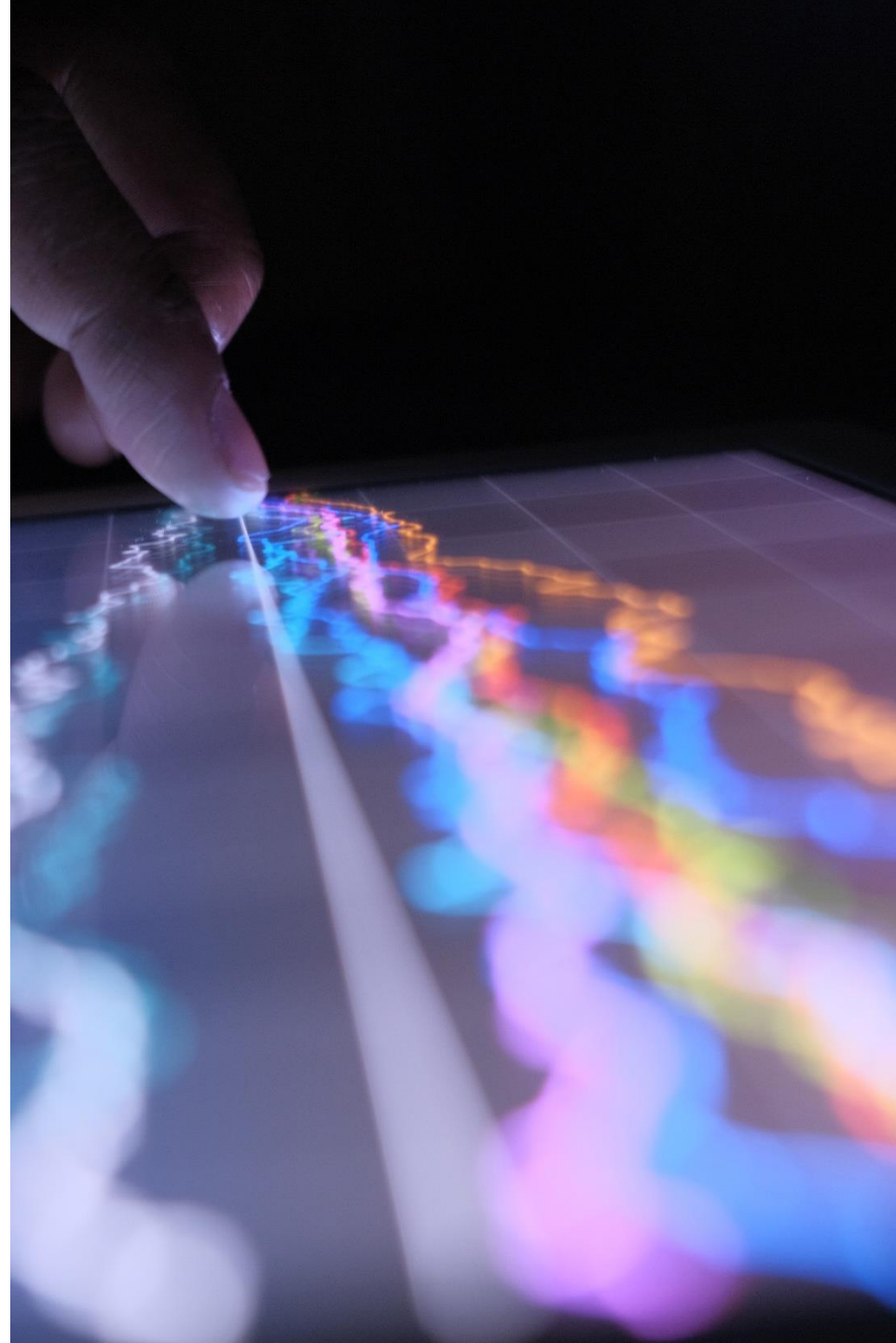
Unit Learning Outcomes

1. Critically evaluate data visualisations and propose improvements based on an understanding of human perception and cognition and data visualisation design principles.
2. Apply a structured design process to create effective visualisations.



Unit Learning Outcomes

3. Conceptualise and iterate data visualisation designs using sketching and low fidelity prototyping techniques.
4. Create web-based interactive data visualisations using a real-world data set.



Learning and Teaching Activities

Live-On-Line (1hr)

- Overview of concepts

Recorded Videos (1-2hrs)

- In-depth concepts

Classes (2 hrs)

- Data wrangling
- Chart design
- Programming
- Project work

Self study and project work (6-8 hrs)



Readings Summary

Data and Data Processing

- Data Mining: Concepts and Techniques: Ch 2&3
- Data Points: Ch 3&4, 5&6
- KNIME Tutorials
- Practical Data Privacy: Ch 1

D3 Programming

- D3.js in Action: Ch 1-3, 4-6, 7, 11-13

Data Visualisation Design

- Visualise This: Ch 3
- Storytelling with Data: Ch 1 & 2



Prerequisites and helpful knowledge

Official

- COS10009 Intro to Programming

Helpful

Web programming:

- COS10005
- COS10026

Data processing, data types, measures of central tendency

- COS10022

User experience design and usability evaluation

- ICT20025



Assessment

Exercise Demonstrations 15% (individual)

Assignment 10% (individual)

Data Visualisation Project 70%

- Stand-ups (15% - individual)
- Visualisation design (20% - team)
- Website (20% - team)
- Reflection and GenAI declaration (15% - individual)



COS30045 GenAI

(Almost) no restrictions on using GenAI

Show understanding:

- Explain work and interactions with GenAI (AI Declaration Report)
- Incremental submissions (e.g., submitting work in progress, on Canvas and/or using versioning control such as GitHub)
- 'Face-to-face' on-campus assessment (e.g., code demonstrations, Q&A, stand-ups)
 - No GenAI
 - Do not rely on comments to get through



GenAI Ethical Issues

Misconduct

- We give the degree to you, not to AI

Attribution

- Giving credit to previous work (not always evident where material has come from)

Code reuse and licencing

- May give you code that you don't have the right to use

Sustainability

- Uses large amount of energy to run AI



GenAI Quality Issues

Training code may not follow best practice

- Produces old code practice

Harmful bias

- Training data may have bias so don't get good representation in

Security

- May produce code that is not secure or has security issues



On-Campus Exercise Demonstration

Submit work and work in progress to Canvas

Include GitHub link

Assessed by **on-campus face-to-face** demonstration

- Explain solution and associated concepts
- Explain the process of how you arrived at solution (including the use of GenAI)
- Update solution to account for new requirement



On-Campus Project Stand-ups

Similar to Exercise demonstrations, but on Project.

Stand-Up 1:

- Data exploration: Audience, purpose and questions

Stand-Up 2:

- Data explanation: Visualisation design

Stand-Up 3:

- Programming: Interactive visualisation

Stand-Up 4:

- Demonstration



Must participate in on-campus assessment

Students who achieve **two or more fail marks** on the Project Stand-Ups may have their **mark reduced** relative to the achieved mark.

Lack of continuity between final Project and works submitted for stand-ups may result in an integrity investigation and/or a **reduction in your Project mark** relative to achieved mark.



Schedule

